

Token-Level vs. Continuous-State Intermediate Computation

DataForge 2026: Pathway Track • NeurIPS 2026 Education Track Format • Topic #6 (Approved) • Word Count: 742 words

The Core Falsifiable Claim: Forcing reasoning models to materialize intermediate deductions as discrete autoregressive tokens imposes an artificial $O(K \cdot L)$ computation and memory barrier that continuous latent state relaxation $z_{\tau+1} = f_{\theta}(z_{\tau}, c)$ accelerates by up to 12× while eliminating token-allocation overhead.

1. The Token Bottleneck in Modern LLM Reasoning

Modern autoregressive large language models perform multi-step deduction via 'Chain of Thought' (CoT). While CoT boosts benchmark performance by allowing the model to allocate additional sequential forward passes, it suffers from a fundamental structural flaw: *intermediate thoughts must be projected into a discrete token vocabulary V at every single reasoning step.*

If a 14-step deduction requires 450 verbal tokens ('Let's first calculate...', 'Notice that...'), the model must perform 450 distinct autoregressive decoding iterations. Each step incurs memory-bandwidth bottlenecks from loading entire model weights from HBM to SRAM and growing the KV-cache by 450 slots (Gu & Dao, 2023; Goyal & Bengio, 2024). This creates a prohibitive inference cost of $O(K \cdot L)$ memory access per query.

2. Continuous Latent Flow vs. Discrete Token Emission

Continuous-state intermediate computation replaces discrete token generation with a continuous dynamical trajectory in latent representation space $\mathbb{R}^d$. Instead of generating tokens $y_1, y_2, \dots, y_K$, the system executes K recurrent micro-steps inside its hidden manifold:

$$z_{\tau+1} = \text{LayerNorm}(z_{\tau} + f_{\theta}(z_{\tau}, c)), \text{ for } \tau = 0, 1, \dots, K-1$$

Why does this work? Reasoning in continuous vector spaces allows the model to explore superposition states and continuous gradient interpolations that discrete token decoders cannot represent. In recent controlled benchmark evaluations (Goyal et al., 2024; Behrouz et al., 2024), continuous latent reasoning achieved equivalent multi-hop task accuracy to 32-step verbal CoT while reducing wall-clock latency by **8.4×** and total GPU energy consumption by **11.7×**.

3. The BDH CQ Frontier: Learning Without Chains of Thought

This exact paradigm is the foundational architecture of Pathway's **BDH CQ** (Pathway, 2026). In the Dragon Hatchling family, BDH replaces discrete autoregressive decoding during intermediate inference with multi-scale recurrent state updates.

When presented with few-shot demonstrations or complex algorithmic constraints, BDH CQ modifies its internal synaptic state $W_t = \lambda W_{t-1} + \eta (y_t \otimes x_t^T)$ and unrolls its continuous latent trajectory z_{τ} across time without emitting a single token into the output sequence. The model outputs only the final answer once the latent trajectory has stabilized at an attractor fixed-point. By decoupling reasoning depth from token count, BDH CQ achieves $O(1)$ memory allocation per reasoning step.

4. Limitations and Open Questions

Continuous latent reasoning is not without severe trade-offs. We identify two primary failure modes:

- 1. Zero Intermediate Interpretability:** Because intermediate computation occurs entirely within continuous vector activations, humans cannot inspect 'what the model was thinking' during steps $\tau = 1 \dots K-1$.
- 2. Attractor Drift and Error Accumulation:** Over deep unrolling horizons ($K > 50$), continuous state updates can drift away from valid manifold regions unless strong regularizers or bounded spectral constraints ($\rho(A) \leq 1$) are strictly enforced during training.

5. The Road Ahead

The belief that intelligent reasoning requires emitting tokens in human natural language is an artifact of autoregressive LLM pretraining, not a computational necessity. As architectures like Dragon Hatchling (BDH) and BDH CQ demonstrate, the future of efficient reasoning lies in continuous latent state dynamics.

Primary References Cited:

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